Companion XXXVIII: The Circumgalactic Medium in the Relaxation-Driven Cyclic Cosmology

Michael Lehmann

mi.lehmann@gmx.de

April 2026

Abstract

The circumgalactic medium (CGM) forms the interface between galaxies and the cosmic web. It regulates gas inflows, outflows, and the long-term evolution of galaxies. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the thermodynamic structure, kinematics, and ionization state of the CGM. This Companion develops a continuous description of the RDCC circumgalactic medium, deriving how the relaxation scale influences the density and temperature profiles, the stability of gaseous halos, and the dynamics of inflows and outflows. The resulting framework provides a natural explanation for the observed coherence of CGM absorption systems and the extended gaseous envelopes around massive galaxies.

1 Introduction

The circumgalactic medium is a vast reservoir of gas that surrounds galaxies and mediates the exchange of mass, energy, and metals between galaxies and their environment. Observations reveal a multiphase structure extending hundreds of kiloparsecs, with cold filaments, warm ionized gas, and hot virialized atmospheres coexisting in a delicate balance.

In the standard cosmological model, the CGM is shaped by the interplay between gravitational collapse, radiative cooling, and feedback. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the growth of small-scale modes and enhances the coherence of large-scale modes. This modifies the density and temperature structure of the CGM, the stability of gaseous halos, and the dynamics of inflows and outflows. The CGM becomes a direct tracer of the relaxation scale $k_{\rm rel}$.

2 The RDCC Potential and the Structure of Gaseous Halos

The gravitational potential that shapes the CGM is modified in RDCC according to

$$\Phi_{\rm RDCC}(k,a) = \Phi(k,a)\left[1 - \chi_\Phi \left(\frac{k}{k_{\rm rel}}\right)^\alpha\right]. \tag{1}$$

This modification reduces the depth of the potential wells of low-mass halos while preserving the coherence of the potential in massive halos. The hydrostatic equilibrium condition,

$$\frac{1}{\rho}\frac{dP}{dr} = -\frac{d\Phi_{\rm RDCC}}{dr}, \tag{2}$$

then yields a density profile that is smoother and more extended than in the standard cosmological model.

The temperature profile follows from the ideal-gas relation,

$$T_{\rm RDCC}(r) = \frac{\mu m_p}{k_B} \frac{1}{\rho(r)} \int_r^\infty \rho(r') \, \frac{d\Phi_{\rm RDCC}}{dr'} \, dr'. \tag{3}$$

The resulting CGM is less centrally concentrated and exhibits a broader transition between the inner halo and the outer gaseous envelope.

3 Cooling, Condensation, and the Multiphase Structure

The CGM is inherently multiphase, with cold clouds embedded in a hot background. The formation of these cold clouds depends on the competition between radiative cooling and gravitational compression. In RDCC, the cooling time becomes

$$t_{\rm cool,RDCC} = \frac{3 k_B T_{\rm RDCC}}{n_e \Lambda(T_{\rm RDCC})}, \tag{4}$$

where the modified temperature profile delays cooling in low-mass halos and accelerates it in massive halos.

The condition for thermal instability,

$$\frac{d \ln \Lambda}{d \ln T} < 2, \tag{5}$$

is reached at different radii in RDCC than in the standard cosmological model. Massive halos develop extended regions of thermal instability, leading to the formation of cold filaments that penetrate deep into the halo. Low-mass halos, whose potentials are weakened by the relaxation mechanism, suppress the formation of cold clouds altogether.

This asymmetry provides a natural explanation for the observed scarcity of cold gas around dwarf galaxies and the prevalence of cold filaments around massive galaxies.

4 Inflow and Outflow Dynamics

Gas flows into and out of galaxies through the CGM. In RDCC, the dynamics of these flows are modified by the relaxation-modified potential. The inflow velocity can be approximated as

$$v_{\rm in,RDCC}(r) = \sqrt{\frac{2 G M_{\rm RDCC}(r)}{r}}, \tag{6}$$

where the reduced mass profile of low-mass halos slows inflows and the enhanced coherence of massive halos accelerates them.

Outflows driven by stellar or AGN feedback must overcome the escape velocity,

$$v_{\rm esc,RDCC}(r) = \sqrt{2 |\Phi_{\rm RDCC}(r)|}. \tag{7}$$

Because the potential is shallower in low-mass halos, outflows escape more easily, leading to a CGM that is metal-poor and diffuse. Massive halos retain their outflows, enriching their CGM and promoting the formation of multiphase structures.

The CGM thus becomes a dynamical record of the relaxation scale.

5 Ionization Structure and Observational Signatures

The ionization state of the CGM is sensitive to its density and temperature structure. In RDCC, the modified profiles lead to characteristic absorption signatures. The column density of an ion species X becomes

$$N_{X,\mathrm{RDCC}} = \int n_X(r)\, dr, \tag{8}$$

where $n_X(r)$ reflects the RDCC-modified density and temperature.

Massive halos exhibit extended absorption systems with coherent velocity structures, while low-mass halos show weaker and more fragmented absorption. These predictions align with observations of Lyman-α absorbers and metal lines around galaxies across cosmic time.

6 Conclusion

The circumgalactic medium in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the density and temperature structure of gaseous halos, the formation of multiphase gas, and the dynamics of inflows and outflows. These effects produce a CGM that is smoother, more coherent, and more extended around massive galaxies, while suppressing cold gas around low-mass galaxies. The present Companion establishes the theoretical foundation for future studies of galaxy evolution, chemical enrichment, and the interaction between galaxies and the cosmic web in RDCC.

References

[1] Lehmann, M. (2026). Relaxation-Driven Cyclic Cosmology (RDCC): A Minimal CPT-Symmetric Framework with a Single Parameter. Flagship Paper. Zenodo: `https://zenodo.org/records/19744958` (v43.0 or later).

[2] The complete RDCC companion series (I–LVII) is available at the same Zenodo record above.

[3] Companion XVI: Boltzmann Code and Modified Hierarchy.

[4] Companion XXXI: RDCC Halo Model and Non-Linear Structure.

[5] Companion XXXII–XXXIX: Galaxy–Halo Connection, Cosmic Web, Lensing, RSD, ISW, tSZ, Rotation Curves, CGM.

[6] Planck Collaboration (2020), Astron. Astrophys. 641, A6.

[7] DESI Collaboration (2024/2025), arXiv: [aktuelle $f\sigma_8$/BAO-Paper].

[8] Maiolino et al. (2024), Nature (JWST high-z galaxies/quasars).

[9] Kaiser, N. (1987), Mon. Not. R. Astron. Soc. 227, 1 (RSD).

[10] Bartelmann, M. & Schneider, P. (2001), Phys. Rep. 340, 291 (Weak Lensing Review).

[11] Hu, W. & Okamoto, T. (2002), Astrophys. J. 574, 566 (CMB Lensing).

[12] Lewis, A. & Challinor, A. (2006), Phys. Rep. 429, 1 (CMB Lensing Review).

[13] Navarro, J. F., Frenk, C. S. & White, S. D. M. (1997), Astrophys. J. 490, 493 (Halo Profiles).